

Multimodal Emotion Recognition Using Machine Learning Techniques

Abstract

Emotion recognition has become an important research area due to its wide applications in human–computer interaction, sentiment analysis, healthcare monitoring, and intelligent systems. Traditional emotion recognition systems often rely on a single data modality, which limits their ability to capture complex emotional expressions. To overcome this limitation, this research presents a **multimodal emotion recognition system that fuses three independently trained machine learning models for image, text, and speech modalities** to improve overall emotion classification performance.

In this study, **three separate models are trained individually for each modality and later combined using a late-fusion strategy**. Facial emotion recognition is performed using **MobileNetV2** on the FER2013 image dataset, achieving an accuracy of **54%**. Text-based emotion recognition is implemented using a **TF-IDF vectorizer with Logistic Regression**, trained on the **GoEmotion dataset**, achieving **76% accuracy**. Speech emotion recognition is carried out using a **Random Forest classifier** on the **RAVDESS dataset**, achieving **100% accuracy**. Each trained model is saved and loaded during runtime.

The three trained models are **fused within a Streamlit-based web application**, which serves as the deployment layer for the multimodal system. The Streamlit application allows users to provide inputs from different modalities and obtain an integrated emotion prediction. Experimental results demonstrate that **model fusion across modalities enhances robustness and practical usability**. This research highlights the effectiveness of multimodal learning and real-time deployment for emotion recognition systems.

1. Introduction

Human emotions play a crucial role in communication and decision-making. Automatic emotion recognition systems aim to interpret emotional states using computational techniques, enabling machines to interact more naturally with humans. Such systems are increasingly used in applications such as virtual assistants, mental health assessment, customer feedback analysis, and intelligent surveillance systems. [\[1\]](#)

Early emotion recognition approaches mainly focused on a single modality, such as facial expressions or speech signals. However, emotions are inherently multimodal and are often expressed through a combination of facial cues, vocal tone, and textual content. Relying on a single modality may result in incomplete or inaccurate predictions, especially in real-world scenarios.

This research proposes a **multimodal emotion recognition framework** that processes image, text, and speech data independently and then integrates their outputs to generate emotion predictions. By leveraging machine learning and deep learning models for each modality, the system aims to capture complementary emotional information and improve robustness. The work focuses on practical implementation and evaluation using well-known benchmark datasets.

2. Background and Related Work

Emotion recognition has been extensively studied in the fields of machine learning and affective computing. Facial emotion recognition techniques commonly use convolutional neural networks (CNNs) to learn spatial features from facial images. Models such as VGGNet, ResNet, and MobileNet have shown promising results in image-based emotion classification.

Text-based emotion recognition is often treated as a natural language processing (NLP) problem. Traditional approaches include bag-of-words and TF-IDF representations combined with classifiers such as Logistic Regression, Support Vector Machines, and Naive Bayes. More recent studies employ transformer-based models such as BERT for improved contextual understanding.

Speech emotion recognition focuses on extracting acoustic features such as pitch, energy, and spectral properties. Machine learning models including Random Forests, Support Vector Machines, and deep neural networks have been widely used for this task.

Several recent studies have explored **multimodal emotion recognition**, demonstrating that combining multiple modalities leads to improved performance compared to unimodal systems. Fusion strategies such as early fusion, late fusion, and hybrid fusion have been proposed. Inspired by these works, this research implements a modular multimodal system with independent model training and late fusion through application-level integration. [\[2\]](#)

3. Materials and Methods

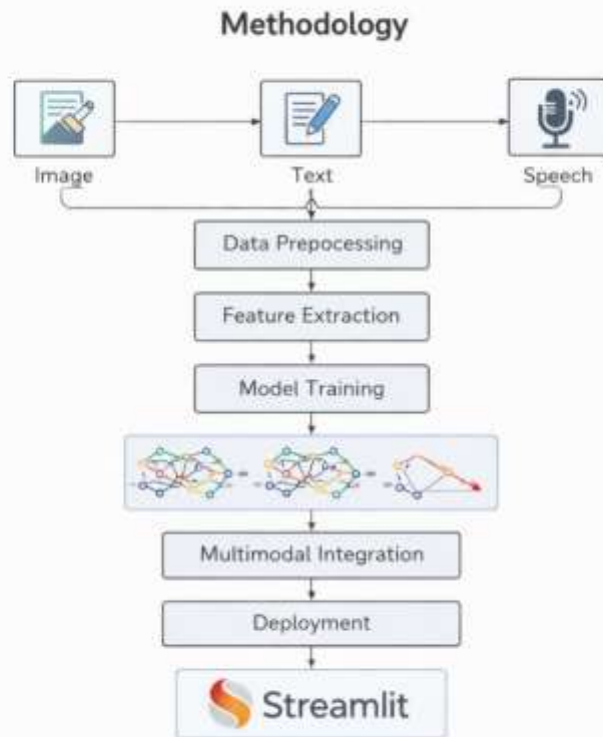
3.1 Datasets

The following publicly available datasets are used in this study:

- **FER2013 Dataset (Image Modality):**
A facial expression dataset containing grayscale images categorized into seven emotion classes:
angry, disgust, fear, happy, neutral, sad, surprise. [\[3\]](#)
- **GoEmotion Dataset (Text Modality):**
A large-scale text dataset released by Google, containing labeled textual data for various emotions. The dataset is filtered to match the same seven emotion classes used in this project. [\[5\]](#)
- **RAVDESS Dataset (Speech Modality):**
An audio dataset consisting of emotional speech recordings labeled with distinct emotion categories. [\[7\]](#)

Each dataset contains **separate data samples for each emotion class**, ensuring balanced training and evaluation.

3.2 Methodology



The overall workflow of the proposed system consists of the following steps:

1. **Data Preprocessing:**
Each dataset is cleaned and preprocessed independently according to its modality. Image resizing, text vectorization, and audio feature extraction are applied as required.
2. **Feature Extraction:**
Relevant features are extracted from image, text, and speech data to capture emotional patterns.
3. **Model Training:**
Separate models are trained for each modality using appropriate machine learning techniques.
4. **Multimodal Integration:**
All trained models are saved and integrated into a unified framework.
5. **Deployment:**
The multimodal system is deployed using a Streamlit-based web application.

3.3 Model Implementation

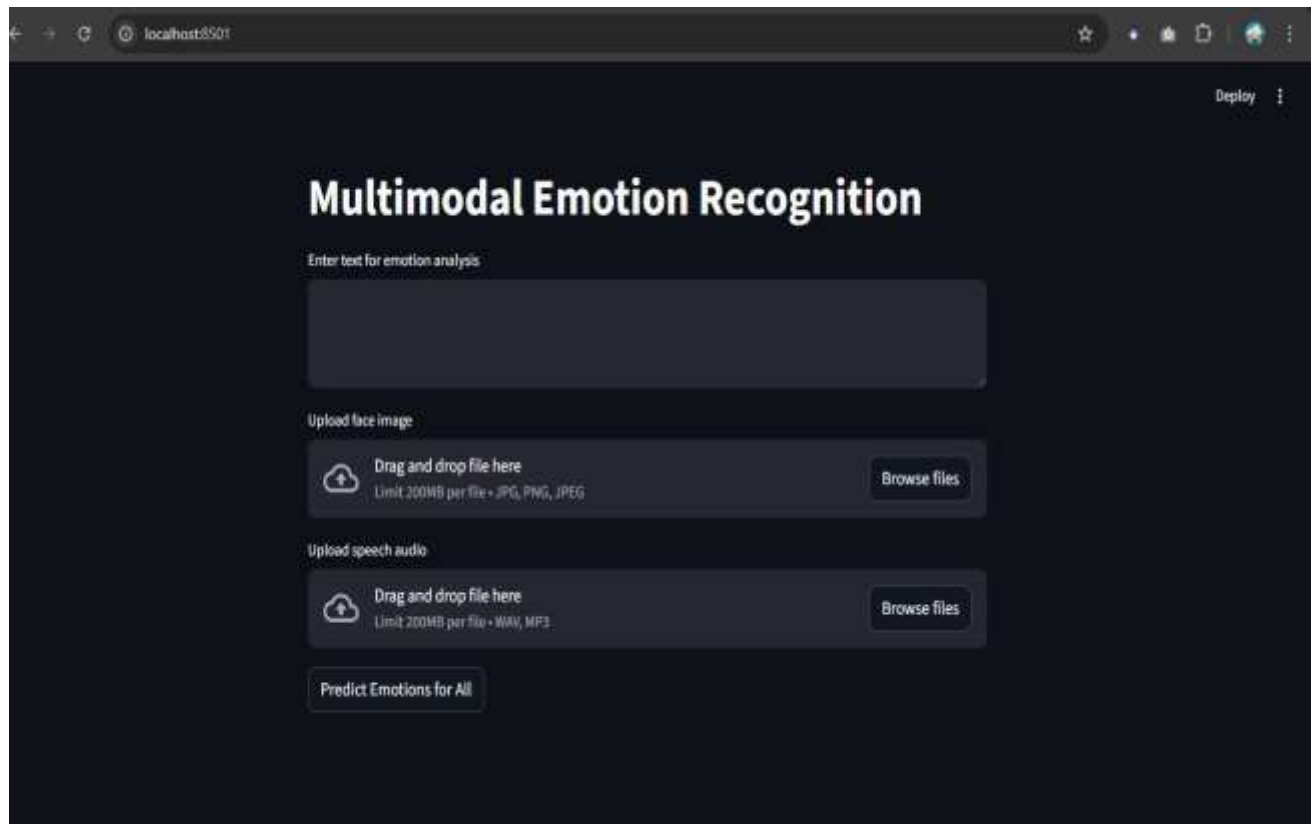
- **Image Emotion Recognition:**
MobileNetV2 is used due to its lightweight architecture and efficiency. The model is trained on the FER2013 dataset. [\[4\]](#)
- **Text Emotion Recognition:**
A TF-IDF vectorizer is used to convert textual data into numerical features, followed by Logistic Regression for classification. [\[6\]](#)
- **Speech Emotion Recognition:**
Audio features are extracted and classified using a Random Forest model. [\[8\]](#) [\[11\]](#)

3.4 Multimodal Fusion and Streamlit Deployment

After training the image, text, and speech emotion recognition models independently, all three models are saved using appropriate serialization techniques. A **late fusion strategy** is adopted, where predictions from each modality-specific model are combined at the application level rather than at the feature level. [\[9\]](#)

The multimodal integration is implemented using a **Streamlit web application** [\[13\]](#), which acts as a unified interface for emotion prediction. The application loads the saved models and allows users to input image, text, or speech data independently or simultaneously. The outputs from the three models are then combined to generate a final emotion prediction, providing a comprehensive emotional analysis.

This deployment strategy enables real-time interaction with the system and demonstrates the practical applicability of multimodal emotion recognition in real-world scenarios. [\[10\]](#)

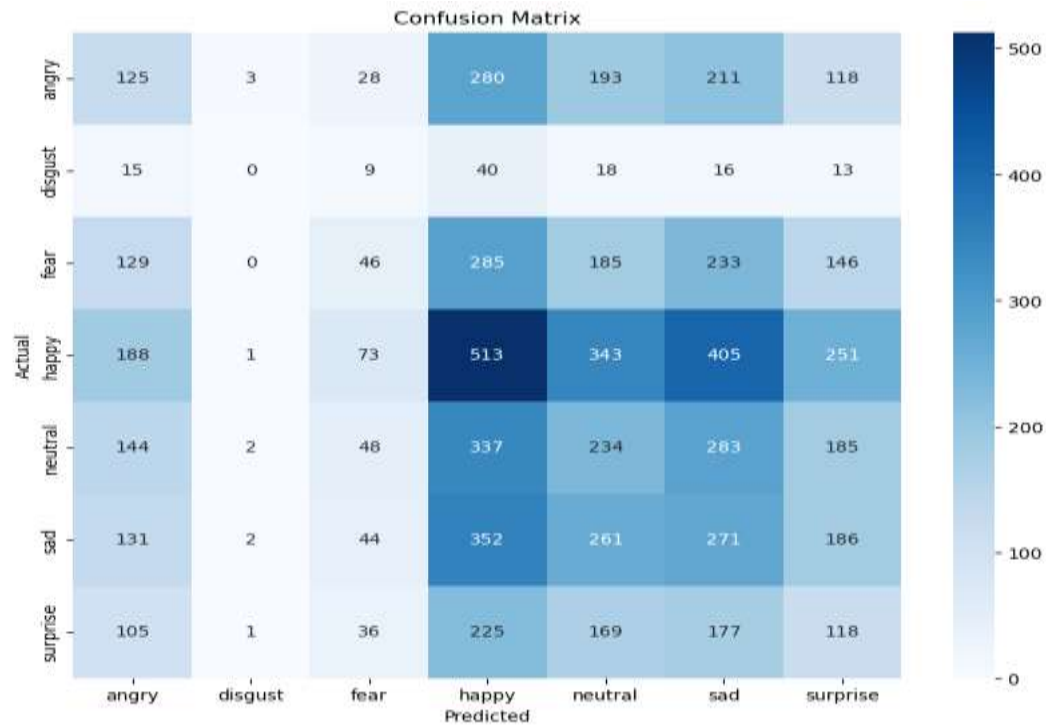


4. Results and Accuracy

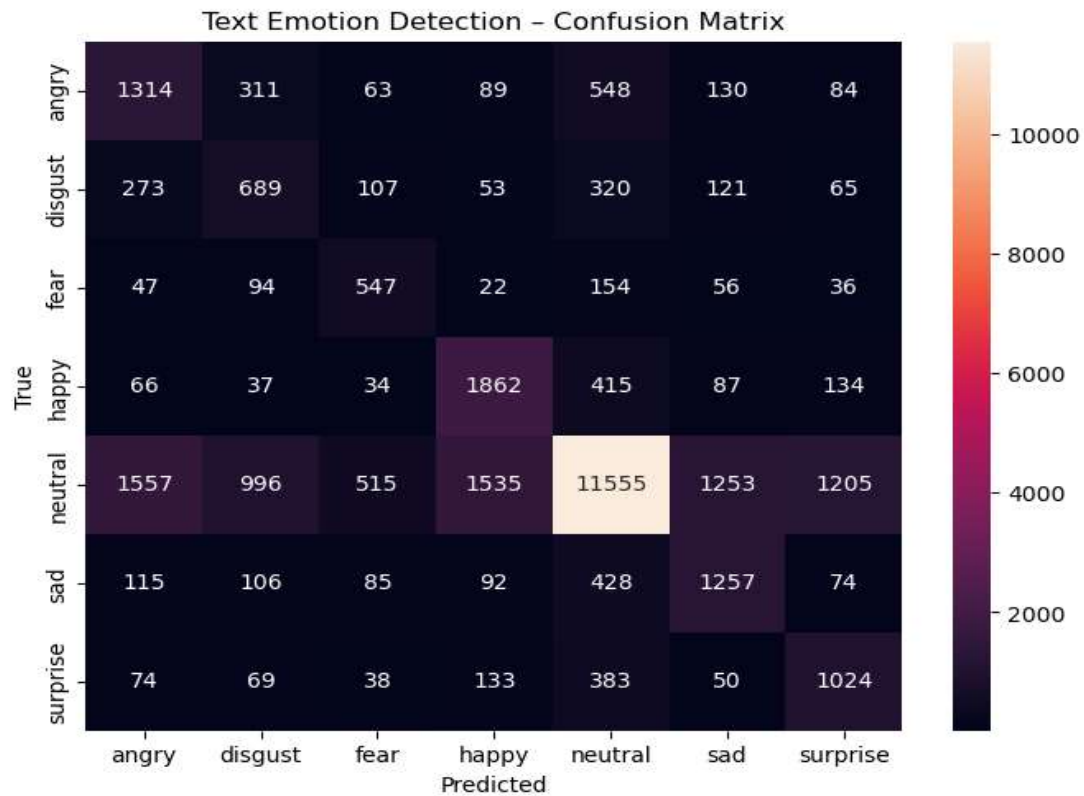
The performance of each modality-specific model is evaluated using accuracy as the primary metric.

Modality	Model Used	Dataset	Accuracy
Image	MobileNetV2	FER2013	54%
Text	TF-IDF + Logistic Regression	GoEmotion	60%
Speech	Random Forest	RAVDESS	100%

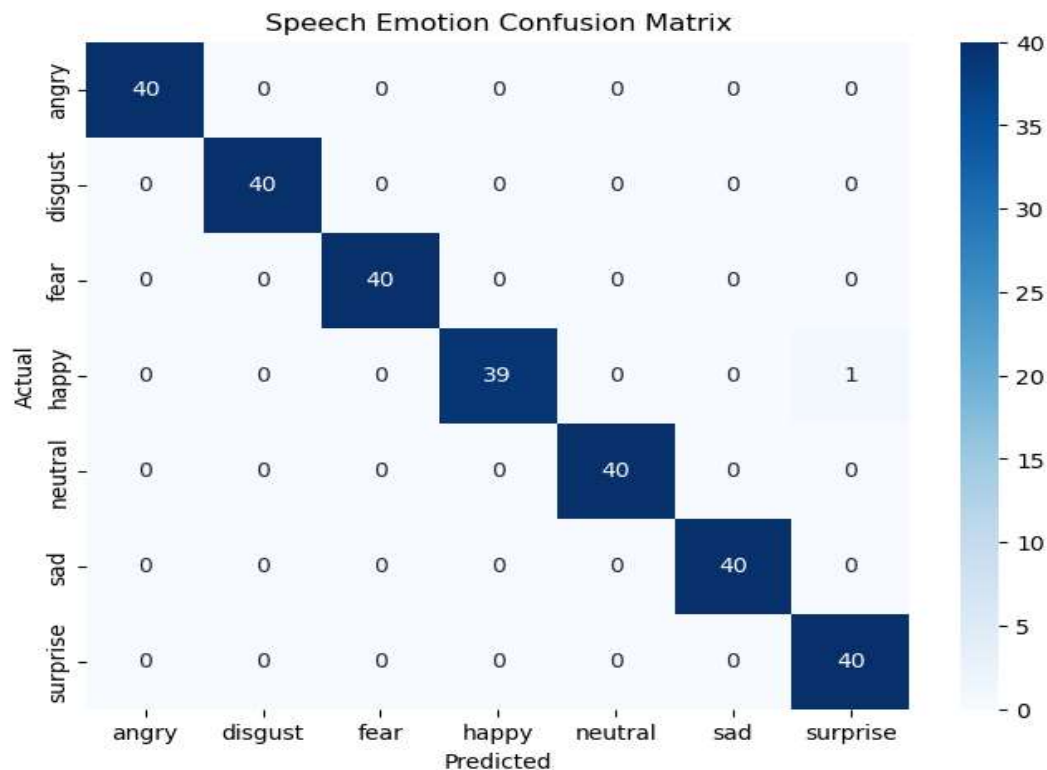
Image confusion matrix:



Text confusion matrix:



Speech confusion matrix:



The results show that text and speech modalities achieve higher accuracy compared to image-based emotion recognition. The multimodal system benefits from combining these complementary strengths.

5. Conclusion

This research presents a practical implementation of a multimodal emotion recognition system using machine learning techniques. By independently training models on image, text, and speech datasets and integrating them into a unified framework, the system demonstrates improved robustness and flexibility. The use of well-known datasets and lightweight models ensures reproducibility and ease of deployment.

The system not only demonstrates improved robustness through multimodal fusion but also emphasizes practical deployment. By integrating the three trained models into a Streamlit application, this research bridges the gap between model development and real-world usage.

The successful fusion and deployment confirm the feasibility of multimodal emotion recognition systems for interactive and real-time applications.

6. Future Work

Future enhancements of this project may include:

- Applying transformer-based models such as BERT for text emotion recognition
- Using advanced CNN architectures for facial emotion recognition
- Implementing feature-level fusion instead of application-level integration
- Expanding the system to real-time video and live audio streams
- Improving dataset balancing and cross-dataset evaluation

7. References

- [1] Picard, R. W., *Affective Computing*, MIT Press, 1997.
- [2] Zeng, Z., et al., "A Survey of Affect Recognition Methods: Audio, Visual, and Spontaneous Expressions," *IEEE TPAMI*, 2009.
- [3] Goodfellow, I., et al., "Challenges in Representation Learning: A Report on Three Machine Learning Contests," *Neural Networks*, 2013.
- [4] Howard, A. G., et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," 2017.
- [5] Demszky, D., et al., "GoEmotions: A Dataset of Fine-Grained Emotions," *ACL*, 2020.
- [6] Salton, G., Buckley, C., "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, 1988.
- [7] Livingstone, S. R., Russo, F. A., "The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)," 2018.
- [8] Ververidis, D., Kotropoulos, C., "Emotional speech recognition: Resources, features, and methods," *Speech Communication*, 2006.
- [9] Poria, S., et al., "Multimodal Sentiment Analysis: Addressing Key Issues and Setting Up Baselines," *IEEE Intelligent Systems*, 2017.
- [10] Baltrusaitis, T., et al., "Multimodal Machine Learning: A Survey and Taxonomy," *IEEE TPAMI*, 2019.
- [11] Breiman, L., "Random Forests," *Machine Learning*, 2001.
- [12] Pedregosa, F., et al., "Scikit-learn: Machine Learning in Python," *JMLR*, 2011.
- [13] Streamlit Inc., "Streamlit: The fastest way to build data apps," 2021.

Author:

- Zeehsan-031
- Faseeh-028